

Agrima Jain

Software Engineer

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PROFILE

Software engineer with 4 years of experience across backend systems, applied machine learning, and full-stack product delivery. Highly experienced in Python services and data pipelines, and has also built C++ and C#/WPF components for a real-time computer vision product, from the inference logic through the desktop tools operators used to run it. Passionate about building software that non-technical users can trust to run reliably in the field. Pragmatic and hands-on, comfortable moving between performance-sensitive code and the interfaces built on top of it.

SKILLS

Languages: Python, C++, C#, TypeScript, SQL

Frameworks: FastAPI, Node.js, NestJS, .NET/WPF, AWS (Lambda, SageMaker, CloudWatch), Docker, Kubernetes, PostgreSQL, Redis

Practices: MVVM, producer/consumer pipelines, asynchronous programming, module and service-level test coverage, REST API design, CI/CD, Git

Others: LLM/RAG pipelines (Pinecone), prompt engineering, LangChain

EXPERIENCE

AI/ML Engineer

Codamatrix, Boston, MA

Sep 2025 – Present

- Integrated RAG pipelines connecting OpenAI and Claude APIs with Pinecone for vector retrieval across 300,000+ clinical documents, powering grounded extraction from unstructured text.
- Designed and deployed FastAPI microservices and REST APIs on AWS, processing 1,000+ healthcare records daily across multiple internal and third-party data sources.
- Extended the production pipeline with a Whisper-based voice transcription service, adding a new data modality to the existing extraction system.
- Presented pipeline design, evaluation results and tradeoffs to clinical, product and engineering stakeholders, translating their feedback into system changes.

AI Engineer Intern

Amwell, Boston, MA

May 2025 – Aug 2025

- Supported a RAG-based recommendation feature using LangChain, tuning prompts and retrieval settings that raised match accuracy by 35%.
- Helped build Node.js and NestJS services on AWS Lambda to connect AI features into the production application.
- Tested an image-prediction pipeline in PyTorch against 50,000+ images, running evaluation experiments that the full-time team used to finish tuning after my internship ended.

Founding Software Engineer

HappyMonk AI Labs, Bengaluru, India

Jan 2021 – Nov 2023

- Optimized the real-time object detection inference path in C++ for a surveillance product processing live camera frames, reaching 93.5% mAP on a YOLOv5 model trained on 300,000+ labeled frames.
- Built a C#/WPF desktop app so operators could monitor live detection feeds and adjust camera parameters, using the MVVM pattern to keep the UI separate from the underlying pipeline state.
- Pulled live frames from networked IP cameras over TCP/IP into a producer/consumer queue that fed the detection and speech-to-text workers asynchronously, keeping the pipeline responsive under bursty load.
- Wrote module-level tests for the C++ detection code and cross-service checks for the FastAPI/Node.js layer wrapping it, catching regressions before each release.
- Served as a founding engineer on a small team, working directly with customers and the hardware and firmware engineers building the cameras and edge devices to scope software requirements, and documenting the detection and transcription APIs for the rest of the team.

EDUCATION

Northeastern University, Boston, MA

Master of Science, Computer Science

Jan 2024 – Dec 2025, GPA 3.9